ZENROUTE: A MOBILE APPLICATION THAT REMOVES COMMUTER HASSLE USING SATELLITE

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APPROVAL SHEET

This research paper entitled:

"ZENROUTE: A MOBILE APPLICATION THAT REMOVES COMMUTER HASSLE USING SATELLITE"

prepared and submitted by Group 2, STEM – 12 – 3, in partial fulfillment of the requirements in Practical Research 2 for the STEM Strand in Senior High School at Our Lady of Fatima University – Quezon City, for School Year 2025–2026, has been examined and is hereby recommended for acceptance and approval.

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DEDICATION

This research is wholeheartedly dedicated to our beloved parents, who/which has been a constant source of love, encouragement, and support.

To our teachers, for their endless guidance and unwavering dedication to our growth and learning.

To our classmates and friends, whose support and companionship has inspired and motivated us throughout this journey.

And above all, to Almighty God, for His boundless love, wisdom, blessings, and guiding us in every step of this endeavor.

This work is a testament to perseverance, faith, and the power of unity.

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Above all, to Almighty God, for granting wisdom, strength, and perseverance to complete this endeavor.

This study would not have been possible without the collective efforts and contributions of all the aforementioned individuals and institutions.

ABSTRACT

This study entitled **"ZENROUTE: A MOBILE APPLICATION THAT REMOVES COMMUTER HASSLE USING SATELLITE"** aimed to develop and evaluate a mobile application designed to assist commuters in identifying the most efficient routes and transportation options using real-time satellite data.

Specifically, it sought to determine the effectiveness of the application in reducing travel time, improving commuters convenience, and enhancing overall user satisfaction.

The research employed the descriptive research design and involved 50 participants/respondents, who were selected through simple random sampling.

Data were collected using structured questionnaires and GPS logs that are built into the application, and analyzed through weighted mean to determine levels of effectiveness and satisfaction, and paired t-test to compare the predicted and actual travel times generated by the application.

The results revealed that there is a significant difference between the estimated time arrival (ETA) and the actual travel times.

It is concluded that the integration of satellite technology in mobile applications can enhance commuting efficiency and convenience..

The researchers recommend further improvement of the application's GPS performance and compatibility across 5.0 android to later versions to ensure broader accessibility.

Keywords: Satellite Technology, Mobile Application, Commuter Convenience, Real-Time Tracking, Route Optimization.

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CHAPTER I

INTRODUCTION

1.1 The Problem and Its Background

Commuting is one of the students, workers, and an everyday need for everyone who does not have their own vehicle. It gives them the privilege to get them from one point to another. Without a vehicle to ride, they would most likely be late to occasions or important events.

Different factors contribute on why commuting can be a hassle for everyone. According to Burlison (2015), It can be challenging because of the time it takes to get to another point. Another factor is where you do not know if the vehicle is full whilst being in a rush.

Factors like full vehicles, route changes, and delays often cause stress and blunders the daily routines, and lowers people's productivity. This inconsistency of public transport has mostly affected people to be late to their classes, work, or meeting places. This application however might lessen, or could help both transit operators and passengers.

This study aims to explore the advancement and impact of an application that uses satellites to track drivers and their routes, similar to booking apps like grab and indrive. This, however, utilizes public vehicles instead of private cars that commuters book to get to another point. This utilizes satellites to track incoming PUV's. By using GPS, the app will deliver real-time location and route information to commuters (Jayathama et al., 2024). This significantly reduces uncertainty and improves convenience. This causes a more organized and much safer commuting experience (AbdelAziz et al., 2019).

1.2 Statement of the Problem

This study aims on how to use satellites on an application to provide real-time detection of incoming public vehicles in order to improve convenience for commuting. This also aims to utilize the ISO/IEC 2501n specifically 25010 where the researchers utilize its reliability, interaction capability, and maintainability where it lessens fault whilst doing a special action,

makes it operational and available for all, to give the users the deserved quality, and to make it useable for everyone with less complications whilst maintaining its functions. For it to be compatible with devices, the researchers would lessen the app graphics and optimize to a certain version that could be used in every device from android 5.0 to later versions. Another is to give the estimated time of arrival of the vehicle that could have an idea on whether the vehicle is near or far. The use of this significantly reduces the waiting time of the commuters as well as the peace of mind to ride a vehicle.

1.2.1 Specific Sop's

1. How can the ETA be integrated successfully in a mobile application?

2. How consistent is the application's GPS tracking while moving across different areas with weak to none signal strengths?

3. How can ISO/IEC 25010 specifically the reliability, interaction capability, and maintainability be implemented successfully in the application?

4. How can we apply the compatibility to systems that have version 5.0 to later versions.

1.3 General Objectives

The main objective of this study is to test the effectiveness of ZenRoute in reducing commuter waiting time. The researchers aim to develop a mobile application that utilizes Public Utilities Vehicles (PUVs). Specifically, this study uses satellite technology to accurately track vehicle locations and provide commuters with real-time information of estimated time of arrival (ETA). This study seeks to improve the commuting experience by acknowledging passengers to plan their travel efficiently and arrive at the designated stop at the appropriate time. Additionally, the researchers intended to improve convenience in transportations, minimize the waiting periods, and promote more effective, organized and time-efficient commuting systems.

Moreover, this study aims to implement the ISO/IEC 25010 software quality standard, aiming on the key attributes such as reliability, interaction capability, and maintainability, which aims to lessen the faults during operation, ensure maximum consistent performance and maintain overall functionality and availability of the application to all users. This study seeks to ensure compatibility with older versions of the Android operating system, specifically from Android 5.0 and later versions to promote wider accessibility. Through achieving this objective the application will improve and provide greater convenience for commuters and provide cost effective reliable solutions for daily transportation needs. To test the effectiveness of ZenRoute in reducing commuter waiting time, the specific objectives are the following:

1.3.1 Specific Objectives

1. To integrate an effective system to calculate the ETA of the incoming vehicles.
2. To make it compatible with devices with version 5.0 to later versions.
3. To evaluate the reliability of the application's GPS tracking system while moving in various signal environments.
4. To implement the ISO/IEC 25010, specifically the reliability, interaction capability, and maintainability, in the development of the application to ensure the dependability of the application.

1.4 Significance of the Study

This study determined the effectiveness and efficiency of the mobile application ZenRoute which uses satellite technology to reduce commuter hassle. Public Utility Vehicles (PUV) are tracked in real time, and the application gives estimated times of arrivals. Additionally, this study will benefit the following:

For Commuters - ZenRoute empowers commuters to plan and adjust commuters' schedules effortlessly by knowing precisely when a ride will arrive. Real-time updates enable people to have better transportation and minimize the time that commuters spend waiting by the roadside.

For Public Transport Operators - ZenRoute gives operators the possibilities to monitor the vehicle's movements in real time, control the schedules, and improve the customer service by eliminating service gaps.

For Government and Local Authorities - This study serves as a basis or opportunity for modernizing public transportation systems, aligning with digital transformation in urban mobility.

For App Developers - This research is expected to inspire the public transport app developers to make better use of GPS and satellite technologies in designing and developing the public transport apps.

For Future Researchers – This research can be a basis for further research concerning public transport technologies and how information technology can improve public transport systems.

1.5 Scope and Delimitations

This study focuses on the development of an application designed for Public Utility Vehicle (PUV) commuters to make traveling easier and more convenient. The app, called ZenRoute, works like Google Maps or Waze and provides optimized travel routes using satellite data. It uses real-time GPS tracking to detect vehicles that are coming, which helps lessen waiting time and makes commuting simpler for students, workers, and other passengers. This study is limited to testing ZenRoute's ETA accuracy in urban jeepney routes within Quezon City. It does not also include other types of non-public transportation, like private cars, school buses, company shuttles, and delivery vehicles.

1.6 Conceptual Framework

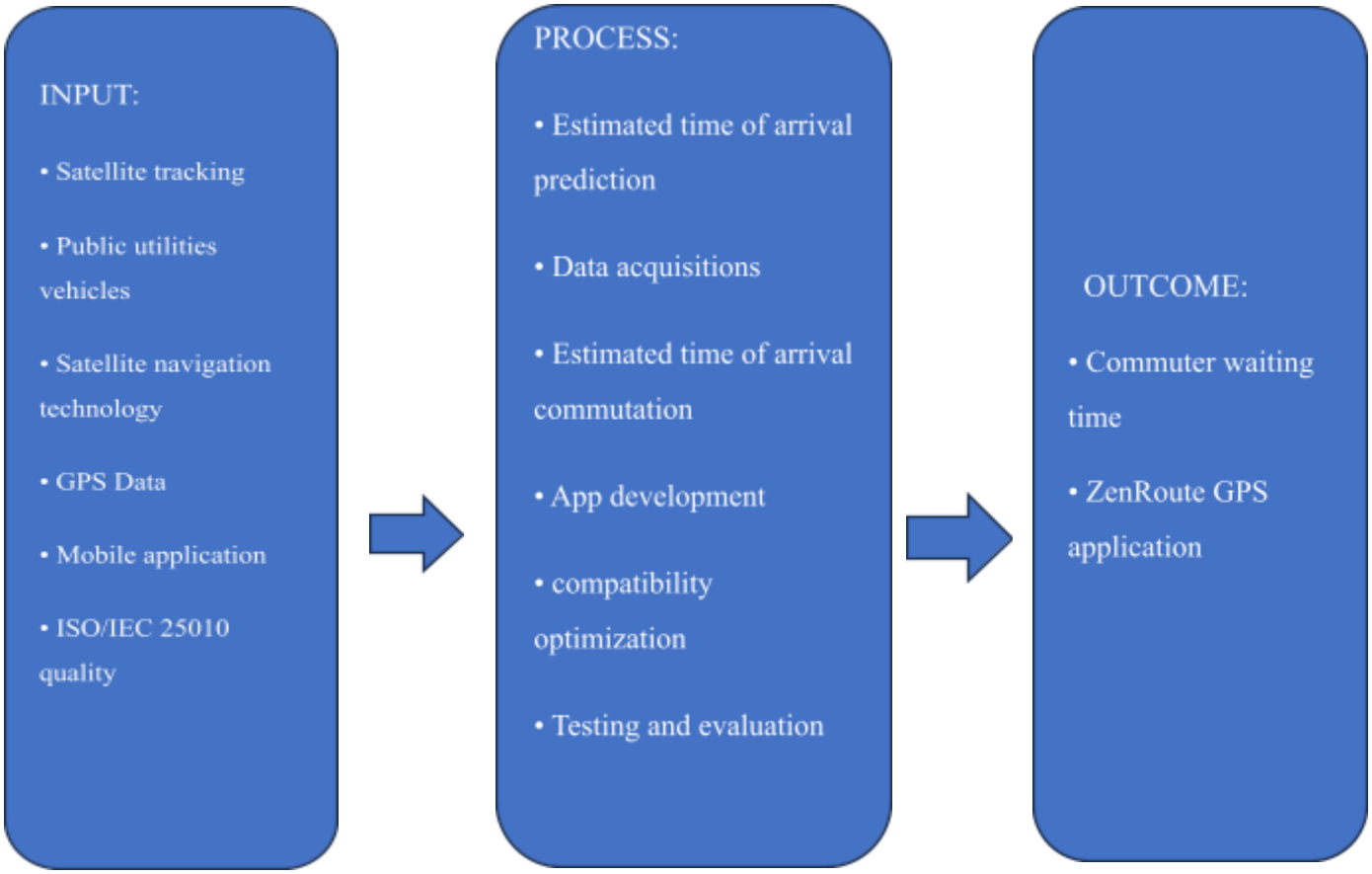

Figure 1. I.P.O. Model

The conceptual framework in this research study follows the Input-Process-Output (IPO) Model. The input consists of Satellite tracking, Public Utility vehicles (PUV), Satellites Navigation Technology, GPS Data, Mobile Application, and ISO/IEC 25010 Quality Standards. The process to create the app comprises Estimated Time of Arrival Prediction, Data Acquisition, Estimated Time of Arrival (ETA) Computation, App Developments, Compatibility Optimization, Testing and Evaluation. The output is the ZenRoute GPS application and the commuters waiting time, this app gives commuters real time locations of Public Utility Vehicles (PUV) using satellite, making their travels much easier as the app makes commuting predictable and transparent.

1.7 Definition of Terms

Application

A program, such as a word processor or a spreadsheet that performs a particular task or set of tasks (Merriam-Webster, 2025). In this study, application means the software program made to keep track of and show public utility vehicles. It works as a place for people to get transportation information on their phones.

Satellite Navigation

A computer-operated system in a vehicle that uses satellites to show the driver the position of the vehicle and which way it should travel to reach its destination (Collins, 2025). In this study, satellite navigation refers to the use of satellites to determine the geographic position of moving vehicles. It ensures that the tracking system functions accurately regardless of time or weather conditions.

GPS (*Global Positioning System*)

A navigational system using satellite signals to fix the location of a radio receiver on or above the earth's surface (Merriam-Webster, 2025). In this study, GPS stands for the Global Positioning System, which is used to find the exact location of public utility vehicles. It makes it possible to track and map things accurately within the app.

PUVs

A vehicle designed to weave in and out of the city streets to fulfill the crucial role of transporting (LTOPORTAL, 2022). In this study, PUV stands for Public Utility Vehicles like buses, jeepneys, and vans that only run on certain routes. The app is mostly keeping track of these cars.

Real-Time Location

The actual time during which something takes place (Merriam-Webster, 2025). In this study, real-time location means the live and constantly changing position of PUVs as seen by GPS. It lets people who are commuting see how vehicles are moving right now through.

Reliability

The extent to which an experiment, test, or measuring procedure yields the same results on repeated trials (Merriam-Webster, 2025). In this study, reliability means that the app consistently and accurately provides correct location data and updates on time. The app's accuracy and uptime while it is running are used to measure it.

User-interaction

A user interaction is any action a person takes when engaging with a digital platform that is met with a response from the system (Clay, 2025). In this study, means how commuters use the app, such as looking up routes, seeing where vehicles are, and getting notifications. It checks to see how well users can use and benefit from the way the system works.

CHAPTER II

REVIEW OF RELATED LITERATURE

2.1 Related Literature

2.1.1 Local

Information and Communication Technology (ICT) serves as a technological enabler of smart cities that supports development and making significant contributions to various sectors such as governance, transportation, education, safety, tourism, and communication. This paper aims to provide a framework for a mobile application that allows citizens to easily access the essential city services, while also allowing local authorities to collect relevant information through the use of literature reviews and related documents (Cabello et. al, 2021).

2.1.2 Foreign

Public transportation in cities play a massive role in reducing costs, while having mobility and environmentally friendly accessibility alternative to private vehicles. However, it restrains the effectiveness of public transport by not knowing the estimated time of arrival (ETA) of the vehicle, and not knowing if the vehicle is packed. Studies have shown that commuter's satisfaction is heavily influenced by factors such as accessibility, reliability, comfort, safety, and security (Chan et al. 2020). A specific technical innovation, specifically real-time tracking via Global Positioning System (GPS), has surfaced as an effective method to enhance public transport systems. The integration of real-time GPS would lessen uncertainty and provide commuters accurate up-to-date information on public vehicles locations and arrival times.

Urban traffic movement is one of the largest problems during rush hours, particularly in the presence of large parked cars which can jam roads and represent safety hazards. Subburaj et al. (2023) developed a waste collection system depending on dynamic traffic movement, nearness to stops, and capacity of bins to offer the most suitable collection time. It employs a routing algorithm for safe and economically charged waste collection and reducing road traffic congestion. The operation can also be monitored and inspected through a mobile application, as well as offering alerts to public mapping services to update commuters regarding route

obstructions created by waste trucks. This system not only enhances waste collection service, but also minimizes road accidents with less traffic congestion with unnecessary delays during journeys for drivers.

Real-time transit information (RTI) aims at making the commuting experience better by decreasing the amount of waiting time, by increasing perception and satisfaction levels, and by encouraging use of public transport; standards such as GTFS-Realtime are put in place for ensuring that data remains accurate and available. The case of Digital Matatus in Nairobi illustrates how GPS can accurately map informal transport routes, and this approach can also be applied to jeepneys in the Philippines (Brakewood & Watkins, 2019).

Sharma (2017) stated that in New Zealand, ITS proves to be a key consideration in achieving a safe, efficient, and resilient transport network. ITS integrates information and communication technologies to use and optimize infrastructure and vehicles through real-time data sharing between the service provider and the user. Here, the National Land Transport Programme elaborates stating that since the government is committed to innovation in the transport sector, $3.2 billion NZD was allocated by the government for ITS-related infrastructure development and research. The study also traced their previous development and applications, stressing that these innovations will continue to play a vital role in the future of the transport system of New Zealand in terms of sustainability and adaptability.

2.2 Related Study

2.2.1 Local

According to Caberto et. al (2024), Routie was developed as a mobile application designed to ease commuting challenges in the Philippines. A prototype was created and evaluated through a user experience test with 50 participants. The evaluation measured functionality, reliability, and usability, with results showing that accurate location reporting and ETA information received the highest ratings. Findings also highlighted that practical effectiveness outweighed entertainment or hedonic features in commuting applications.

According to Fillone, Tamon, & Torrendon (2024), the study introduces Byahero, a GPS-based mobile application designed to help the commuters in Iloilo City to track public transport routes more effectively. The study shows the effectiveness of using GPS on tracking the routes of public transportation. This study provides specific guidance when it comes to what public transportation will pass the route and the estimated time of arrival (ETA), and the schedules.

2.2.2 Foreign

As stated by Chen et al. In order to improve the accuracy of bus travel time predictions, the study suggests a Bayesian Gaussian mixture model in 2022. The model uses link travel times and headways to account for the relationship between successive buses. Compared to conventional deterministic models, this method is more dependable because it works well even with noisy or missing data. The findings demonstrate that it is capable of offering complete predictive distributions of bus travel times in addition to accurate point estimates. As a result, the system is more beneficial to commuters and operators in terms of trip planning and effective public transportation management.

Hossain et al. (2022) propose SPaFE, a crowdsourced multimodal recommender system to enhance safety for traveling in urban settings. It is to that challenge that the study responds by focusing on the issue of emerging safety problems in cities, interacting for instance for commuters circulating in unknown or unsafe environments. The system employs a novel combination of crowdsourced data, real-time user feedback and multimodal transit information to determine and recommend safer journeys. The finding demonstrates that SPaFE leads to a more effective navigation, which takes into account the factors including distance, time and safety. The authors suggest that such systems may have a crucial function in smart city construction and passengers' protection. They suggest additional model tuning, large scale deployment, and collaboration with popular navigation apps for broader use and application.

The study by Weng et. al (2016) presented a model that estimates bus travel speed in real time using GPS data. The method involved preprocessing GPS points, map-matching them to a GIS road network, and calculating travel speed along route arcs. Validation results showed an

accuracy of 88.4% for segment speeds and 97.9% for line speeds. After optimization, which included adjusting stop locations and increasing GPS frequency, the model achieved 91.4% precision.

2.3 Synthesis of the Review of Literature and Studies

The studies and literature are connected to this research, as these works provide knowledge on how satellites and mobile technology help commuters on navigating their routes. Based on the previous study, it shows that commuters' satisfaction depends on the accessibility, reliability, comfort, safety and security (Chan et al. 2020). Apart from this, decreasing the amount of waiting time is also one of the key factors to increase the commuters' perception and satisfaction levels (Brakewood & Watkins, 2019), which aligned to the goals of ZenRoute: A Mobile Application That Removes Commuter Hassle Using Satellite.

Local studies, including Routie (Caberto et. al., 2024) and Byahero (Fillone, Tamon, & Torrendon, 2024) emphasized the effectiveness of GPS-based mobile applications in tracking public transport routes and offering estimated time of arrival (ETA) in Philippines settings. Similarly, foreign studies by Subburaj et al. (2023), Brakewood and Watkins (2019), Sharma (2017), and Chen et al. (2022) have demonstrated the effectiveness of GPS routes in the development of smart city, showing how intelligent systems enhance transportation efficiency, enhance commuter satisfaction, and improve safety. By synthesizing these studies and literatures, the ZenRoute is a timely and necessary innovation aimed to provide commuters a hassle-free and convenient travel.

CHAPTER III

METHODS, APPROACH AND PROCEDURE

3.1 Methodology

This chapter presents the research methods that will be used to conduct the study. It provides research design, research locale, respondents, research instrument(s), data gathering procedure, statistical treatment of data, and ethical consideration. Furthermore, it discusses the project development stage of the research, system architecture, technical feasibility, user interface design, and testing methods. This research employs a quantitative, quasi-experimental design to compare the Estimated Time of Arrival (ETA) derived from the ZenRoute application with the actual arrival time of public utility vehicles (PUVs). This design will evaluate both the accuracy and reliability of the application, as well as the application's ability to reduce waiting time for commuters. These chosen methodologies were implemented to keep the research process systematic and evidence-driven while maximizing the overall goals of ZenRoute. This framework provides the essential basis for assessing ZenRoute in its roles as both a research study and a functional commuter application.

3.2 Research Design

This study will use a quantitative study approach employing a descriptive research design to assess the impact of a satellite based mobile application on reducing commuter waiting time in Quezon city. The focus will be on measuring the frequency, duration, and statistical relationships between commuter experiences, perceptions, and behaviors, as well as the observed changes in waiting times and efficiency among commuters and transportation stakeholders within the specific context of daily commutes. This research aims to determine if using ZenRoute leads to statistically significant reductions in average commuter waiting times and improvements in overall commute satisfaction. Furthermore, it seeks to quantify the extent to which ZenRoute impacts the efficiency of commutes during peak hours in Quezon City.

3.3 Project Development Methodology

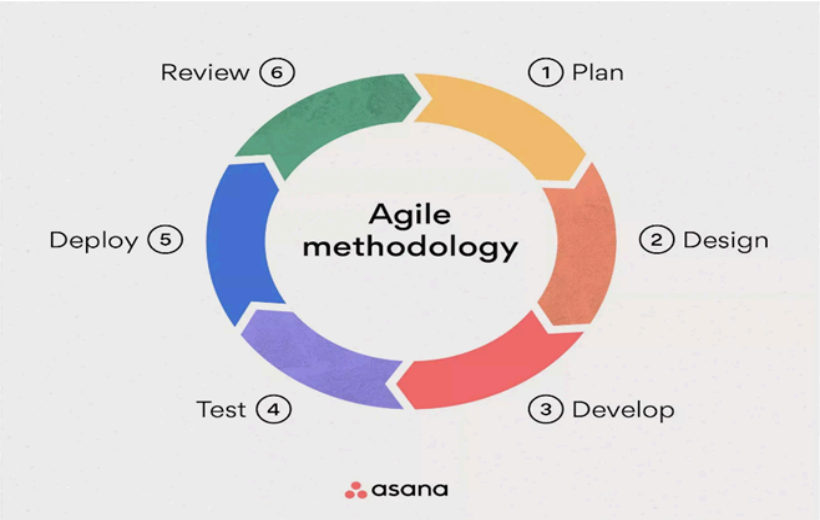

Figure 2. Agile SDLC Model

This project will use agile methodology as its project development methodology. Agile methodology is a framework that breaks projects down into several dynamic processes, known as sprints. This is used because of its adaptability and where it focuses on users. In this project, we have 6 processes where we will have the plan, design, development, testing, deployment and review. This will describe the flow of the project on how it will be done step by step.

Plan

The initial plan for this project is where it will be focused on commuters or people that use public utility vehicles for their daily life. This application is planned to identify real-time traffic, and have GPS-based route tracking. The goal of this project is to minimize the commuter's stress in dealing with public transportation.

Design

This will use a simple UI/UX design where it will be easy to interact with and pleasing to the eye. Another design that this application would utilize is data flow design for its satellite data retrieval and processing. The goal of this part is to help the user to have easier access and have a usable application anytime.

Develop

In this part of the project, this project will use JavaScript as its main programming language as it is the best for application development. This will help build the application from scratch and help encode the plan and design into the mobile application.

Test

In this phase, there will be tests where the application is used to its extent and see if there are any malfunctions and bugs. This phase would help determine bugs and give the best experience for users.

Deploy

This phase is where the application will be deployed and will be publicly available. In this part, it needs to be properly working and would be smoothly running for users. This requires minimal to none bugs as where it would not crash and would not malfunction when deployed publicly.

Review

This part mostly consists of feedback as to where it will ask the user if it encountered any malfunctions or bugs. This also consists of where the developers would address and solve the issues immediately. Additionally, receiving feedback would help the application's growth as to where the upcoming updates would provide a more interactive and smoother experience for the public.

3.4 System Architecture

The Zenroute system uses a client-server architecture with cloud integration. The ZenRoute is installed on the commuter phone by this the app is directly connected to the cloud server through the internet. This architecture supports the experimental testing for ETA tracking, as it enables real-time monitoring of Public Utility Vehicles (PUVs) using GPS and satellite data.

The cloud server processes information on the app and gives the locations of PUV and estimated arrival times, while the route data, GPS history, and commuter details are stored in the cloud database. When a commuter opens the app, the app sends a request to the server cloud. After that, it processes information such as checking the PUVs location and estimating how long before they arrive. Inside the cloud, the cloud server processes the requests and does the ETA calculations. While the cloud database keeps the route data, GPS history, and commuter details.

The ZenRoute is also linked with the satellite and GPS services. This continuously feeds the system with real-time location of the vehicles. The server then updates the app instantly, notifying the user that their ride will arrive. Through this cloud-based setup, ZenRoute is able to keep the commuters constantly connected wherever they go. It also enables the app to handle numerous users simultaneously while maintaining the data as well as timely.

Client Side (Users/Commuters):

- Mobile Application (Android 5.0 and above)
- Sends requests and displays real-time PUV data

Application Server:

- Processes ETA calculations
- Handles GPS and API requests
- Supports *experimental testing for ETA tracking*

Database Server (Cloud):

- Stores routes, Syncs data with satellite and GPS services
- GPS data and user info

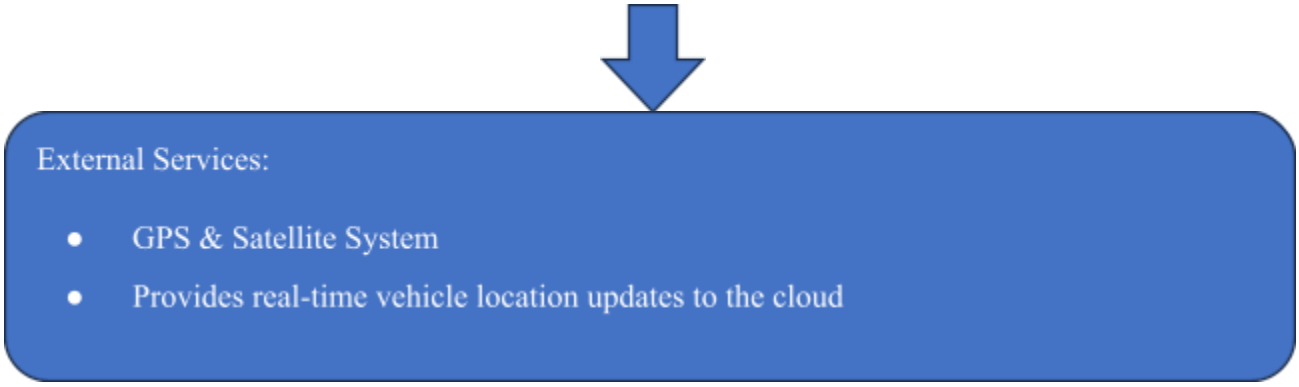

Figure 3. Three-tier client-server architecture model

The ZenRoute app uses a Three-tier client-server architecture model. The client side shows the mobile application used by the commuters to send and receive real-time data. The application server processes ETA calculations, Handles GPS and API testing and it supports experimental testing for ETA tracking. The database server stores all route information, commuters information and GPS services while synchronizing the data with external satellite and GPS services that provide real-time vehicle location. This system ensures accurate, real-time tracking and reliable data exchange between users and the app.

3.5 Technical Feasibility – Hardware/Software Requirements

Hardware:

The proposed system requires basic hardware components to function effectively. Smartphones are designated as the primary devices as these facilitate data transmission between the driver and the commuter. To support a durable internet connectivity, each smartphone must operate with a SIM card as well as the data plan. In addition, a power bank is also needed to provide continuous power supply during long routes. The hardware components are presented in the table below:

Picture	Material	Quantity	Price per Units (PHP)	Purpose
	Smartphones	*2 units*	*10,000*	*Both ends have designated smartphone units to access one other. One unit is designated for the driver to have real-time location data on commuters. On the other hand, the unit is assigned for commuters to have an access and monitor the real-time location data of Public Utility Vehicle (PUV).*
	Sim Card with Data Plan (Globe, Smart)	*2 units*	*300*	*This provides persistent mobile internet connectivity. It allows the smartphones to*

				exchange location and information in real time.
	Power Banks	*1 unit*	*2,500*	*This serves as a power supply, especially to the drivers. It helps to prevent the driver's smartphone from shutting down during long routes.*

Table 1

Software:

The proposed system relies on essential software components to operate efficiently. Each feature is designed to address and reach the commuters' needs. The features of the system includes time-to-time updates on seat availability of PUV, defined route coverage, and monitoring the movement of vehicles. These features of the system deliver accurate real-time information, which improves the efficiency, feasibility, and reliability of commuting. By integrating these features, the system creates a feasible and reliable platform that effectively connects the drivers and the passengers. The software components are presented in the table below:

Software Component	**Description**	**Purpose**
Passenger Capacity Module	*A feature that displays the number of available seats in*	*This presents real-time updates on seat availability*

	the Public Utility Vehicle (PUV).	*that assists commuters in planning trips efficiently.*
Specific Reach Module	*A feature that presents the precise route coverage of the PUV.*	*To provide a clear overview of the designated route and destination, reducing possible confusion in travel.*
Signals for Routes Feature	*A feature that tracks the PUV's movement in real time through GPS and internet connectivity.*	*This provides commuters with accurate information on the vehicle's location to minimize waiting time.*

Table 1.1

3.6 User Interface Design

The researchers make mockups on how the application runs. The first screen when the user opens the application, the user sees the user account creation and login. There are separate mockups for user creation accounts and login. There are a series of steps to create an account, including email and password setup, and phone number verification code. When the account is created the terms and conditions will appear on the screen. In the other button, when you click login, there is some information that the application needs to open the user account. When the user forgot its password, the user can click the button under where to enter the password. The application will transition to a series of mockups, including resetting the password, email verification, and creation of a new password.

Once the user logged in, the application will transition to the main home screen. The screen displays the user's current location in the map. There are options for the user to choose the different modes of transportation such as buses, E-jeeps, and UV express vans. The application suggests a system that helps users to find and navigate public transportation routes. The application home screen also allows the user to directly go to the user profile, search, and

location options through subsequent mockups. The subsequent mockups detail the functionality of search and location options. The application allows the users to search a specific location and the screen will display the nearby routes of public transportation with estimated time of arrival (ETA).

The final screen of the application is the user profile and location confirmation features. The application allows the user to save the frequently used locations. The overall design of the application is a clean and efficient app to help users to their daily commute by providing the actual flow of public transportations and the real-time information.

3.7 Research Locale

The testing will be conducted on jeepney routes in Lagro, Quezon City. Focusing on jeepney routes passing through Lagro, Fairview, Commonwealth, and Anonas areas. This site was selected as it is a commonly used route by students and daily commuters, ensuring an appropriate number of reliable respondents for the study. The researchers consider this route ideal as it reflects real commuting experiences under typical traffic and environmental conditions. The location was chosen for its accessibility, relevance to the study population, and suitability in replicating actual travel circumstances, making it the most practical and effective environment for data collection.

3.8 Respondents

This study included 50 respondents, consisting of 25 commuters and 25 e-jeep drivers. Respondents were selected through simple random sampling, with no restrictions on age or gender, allowing for a variety of perspectives and experiences. The researchers focused on two respondent groups: public transport users and public utility vehicle drivers. Selection was based on daily participation in transportation activities. Inclusion of the two groups enabled a thorough evaluation of the condition of public transport and related issues.

Commuters

Commuters are people who do public transportation on a daily basis- to work, to run an errand, or to perform any other activity. Such individuals can recount experiences on the crowding, fare increases, and travel time concerned.

PUV Drivers

Jeepney and e-jeep drivers having a minimum of 1-year experience will be respondents. They can speak regarding the traffic situation, fuel costs, operation schedule, and passenger safety concerns.

3.9 Research Instrument/s

To collect reliable and comprehensive data the researcher will utilize a structured questionnaire, and GPS logs that are built into the application. The commuter questionnaire will be divided into three major parts focusing on quantitative data. The first part will include a standardized app satisfaction scale designed to assess commuters overall satisfaction levels through numerical ratings, frequency of use metrics, and feature questions that are focused on their experiences with the app with in-app surveys. The second part will be a rating scale that targets specific app features contributing to waiting time reduction, including real time tracking accuracy, predictive arrival times, route optimization effectiveness, and satellite data reliability by using the GPS logs sent to our server and ensuring that it is accurate. Each item will be rated using a numerical scale 1-5 to measure the perceived impact of these features on reducing waiting time. The third part will involve objective measurements of app performance, such as GPS accuracy (in meters) data update frequency (in seconds) and battery consumption (in percentage), collected through automated logging and system analytics.

3.10 Data Gathering Procedure

A letter of request to conduct the study will be prepared. The researchers will collect data to determine if *ZenRoute*, a mobile application based on satellite technology, could minimize the waiting time of commuters. To gather accurate data, the researchers will follow these steps:

Step 1: Record the Estimated Time of Arrival (ETA) from the ZenRoute application for selected routes.

Step 2: Observe and record the actual arrival time of public utility vehicles (PUVs) at the designated pickup points.

Step 3: Compute the difference between the ETA and the actual arrival time to determine the accuracy and efficiency of ZenRoute.

Step 4: Distribute survey questionnaires to commuters and drivers to gather their experiences and perceptions regarding waiting times and the app's usefulness.

Step 5: Collect and compile all responses for data analysis.

The main participants are students, commuters, and drivers of public utility vehicles (PUVs). To ensure everyone has an equal opportunity to be included in the research, the researchers will employ simple random sampling when choosing respondents. A survey will serve as the primary instrument in gathering data. Questions are framed to learn how commuters experience waiting times, and how ZenRoute might reduce waiting times. The researchers will distribute the questionnaires face-to-face to directly reach the participants.

Before answering, participants are informed of the purpose of the study prior to answering and are assured that their responses would remain confidential and their participation is voluntary. The survey has no time limit restriction, in order for the respondents to answer the questionnaire at leisure. Upon completion, the responses gathered are carefully collated and prepared for examination.

3.11 Testing and Evaluation

The Information Technology and Computer Science studies will use Testing and Evaluation, a standardized process to evaluate if a system meets specifications or provides satisfactory user experience. The Testing and Evaluation for ZenRoute will be conducted to determine if the application provides real-time Public Utility Vehicle (PUV) tracking, valid PUV estimated arrival times (ETA) and usability under a variety of conditions. This is an important test due to the reasons already given. Early problems detection and solutions reduces the possibility of operational failures, increases the chance of user adoption, and decreases user experience issues after deployment. Evaluation for ZenRoute will be conducted using two Evaluation methods: The System Usability Scale (SUS) and the ISO/IEC 25010 software quality model as part of the software quality evaluation framework.

The System Usability Scale (SUS) is a recognized standardized 10-item questionnaire to calculate overall usability on the metrics of Learnability, Efficiency, and User Satisfaction. The system calculates a score from 0 to 100, in which 68 and above are capable of achieving above average overall usability. Likewise, the ISO/IEC 25010 framework will be used to assess four key quality attributes of the system: Reliability (the degree of accuracy and consistency of the GPS tracking), Usability (the ease of use and overall user experience), Compatibility (the usability across Android versions 5.0 and above), and Maintainability (the degree of updates and improvements). These evaluation strategies were chosen because they support user feedback through objective quality measures, and therefore will evaluate ZenRoute as user friendly and a high-quality system.

The testing process will include three groups of respondents: students, commuters, and PUV drivers, since they're the main users of public transportation. Participants will be asked to use the application in real and simulated commuting situations. This consists of creation of account setup, locating the vehicles nearby, checking the estimated time of arrival (ETA), and exploring other functionalities. Upon completion, participants will then complete the SUS questionnaire (to collect usability scores, based on "System Usability Scale") and an ISO/IEC 25010-compliant checklist for system quality attributes. The task completion, response time, and number of errors

will also be documented. The outcome from these instruments will then be systematically arranged and subjected to statistical analysis in the succeeding section, ensuring that the evaluation outcomes are interpreted accurately and provide a credible basis for assessing ZenRoute's effectiveness.

In order to achieve results that are both measurable and objective, usability testing for ZenRoute will be conducted using a structured protocol based on the System Usability Scale and ISO/IEC 25010 quality model. It shall focus on important metrics that will assess the user experience and system performance under real and simulated conditions. The following metrics will be evaluated:

Reliability Uptime: It is the percentage of time the system runs without failure during the testing and denotes stability and consistent performance.

Maintainability: The number of functional errors encountered refers to the ease with which the system is maintained or can be updated.

Efficiency: The mean amount of time users spend on important tasks such as establishing an account, locating cars, and determining their estimated time of arrival.

Effectiveness: This refers to the share of tasks that are completed independently, demonstrating usability and intuitiveness.

User Satisfaction: Assessed via the System Usability Scale questionnaire following each testing session, reflecting perceived usability and contentment

Zenroute's Usability and Performance will be measured. Then, results will be analyzed in the light of ISO/IEC 25010 to ensure the overall quality, usability, and dependability of the system.

3.12 Statistical Treatment of Data

In this non-experimental research, the statistical treatment of data is used to organize and analyze information without manipulating variables and will be used to analyze and interpret the data gathered from the respondents. Descriptive statistics such as percentage will be used to describe the demographic profile of the respondents, while the weighted mean will be employed to assess their level of agreement on the reliability, interaction capability, and maintainability of the ZenRoute application.

$$P = \frac{N}{T} \times 100\%$$

Percentage= f / N × 100

$$\frac{\sum(w \times x)}{\sum w}$$

Weighted mean= Σ (w × x) / Σ w

Standard deviation will be used to determine the consistency of the responses and the sample mean will be used to determine the average response of the participants regarding the reliability, interaction capability, and maintainability of the ZenRoute application.

$$s = \sqrt{\frac{\sum(x_i - \bar{x})^2}{n-1}}$$

Standard deviation= √ Σ (x × x)² / N-1

$$\bar{x} = \frac{\sum x_i}{n}$$

Sample mean= Σ xi /n

For hypothesis testing, a Paired t-test will be applied to compare the Estimated Time of Arrival (ETA) predicted by ZenRoute and the actual arrival time of public utility vehicles. This

test was chosen because it compares two related sets of data obtained from the same trips or observations. Since both measurements come from the same set of trips or observations, the paired t-test accurately determines whether there is a significant difference between the predicted and actual times, thereby assessing the accuracy and effectiveness of the ZenRoute system in providing reliable time predictions for commuters. The statistical results will be presented in tables and figures to ensure a clear, organized, and comprehensive interpretation of the findings.

3.13 Ethical Considerations

The ZenRoute study upholds strict ethical standards for the protection of participants: First, confidentiality is achieved by anonymizing personal information. Second, security is maintained by safe storing and limited access to the information. And thirdly, the study maintains the protection of the subjects by not imposing any sort of harm.

The ZenRoute study maintains strict ethical criteria regarding participant protection: Informed consent forms were signed prior to participants' exposure to the experiment. No personal or identifying information regarding the participants was collected; accordingly, privacy and confidentiality were maintained. Potential risks to participants were minimized: secure data handling, voluntary participation, the right to withdraw at any time without penalty.

Participants are explained the purposes of research, procedure, and the data so as to provide an informed and voluntary decision. The gathering of personal data during research is minimal, with the necessary data for research purposes only and treated in confidence. Any data recognized as such is used for research purposes only or de-identified so as not to provide linkage for individuals.

Voluntariness of participation, coupled with the exclusive use of basic and anonymous information, embodies the research's commitment to conducting research conscientiously and responsibly. This approach functions to further enhance participant autonomy and privacy in order that individuals could participate in the research with confidence that their welfare and secrecy would be adequately taken care of.

APPENDIX A

LIST OF TABLES

Picture	Material	Quantity	Price per Units (PHP)	Purpose
	Smartphones	*2 units*	*10,000*	*Both ends have designated smartphone units to access one other. One unit is designated for the driver to have real-time location data on commuters. On the other hand, the unit is assigned for commuters to have an access and monitor the real-time location data of Public Utility Vehicle (PUV).*

	Sim Card with Data Plan (Globe, Smart)	*2 units*	*300*	*This provides persistent mobile internet connectivity. It allows the smartphones to exchange location and information in real time.*
	Power Banks	*1 unit*	*2,500*	*This serves as a power supply, especially to the drivers. It helps to prevent the driver's smartphone from shutting down during long routes.*

Table 1

Software Component	**Description**	**Purpose**
Passenger Capacity Module	*A feature that displays the number of available seats in the Public Utility Vehicle (PUV).*	*This presents real-time updates on seat availability that assists commuters in planning trips efficiently.*
Specific Reach Module	*A feature that presents the precise route coverage of the PUV.*	*To provide a clear overview of the designated route and destination, reducing possible confusion in travel.*
Signals for Routes Feature	*A feature that tracks the PUV's movement in real time through GPS and internet connectivity.*	*This provides commuters with accurate information on the vehicle's location to minimize waiting time.*

Table 1.1

APPENDIX B

LIST OF FIGURES

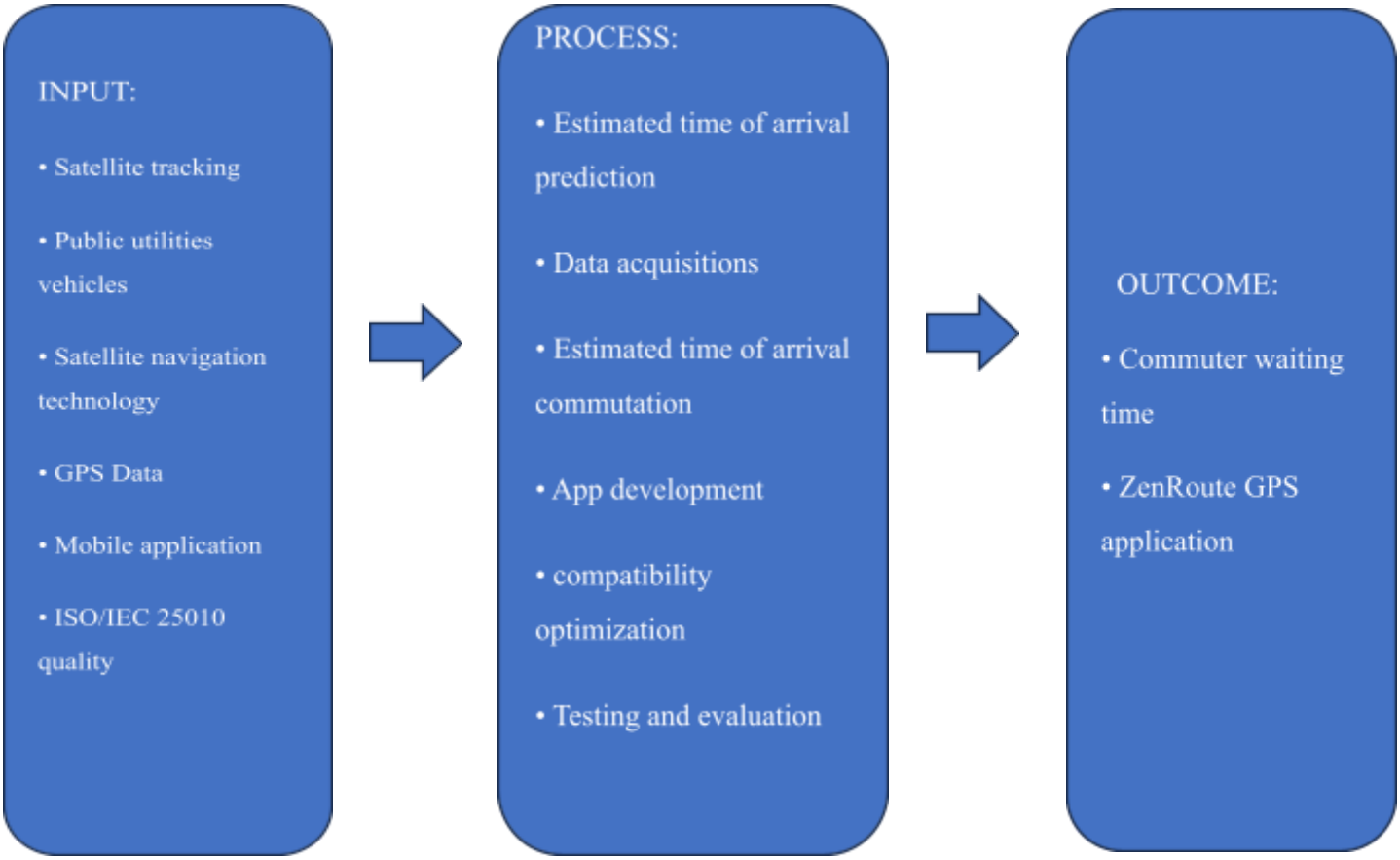

Figure 1. I.P.O. Model

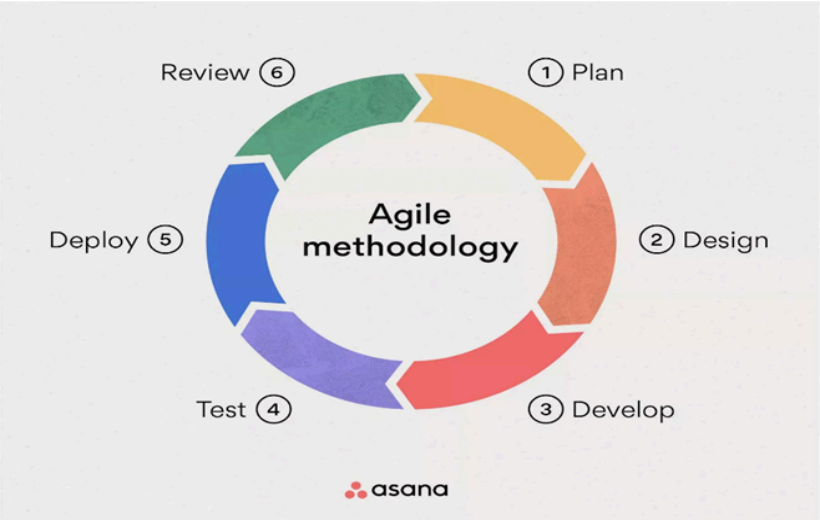

Figure 2. Agile SLDC Model

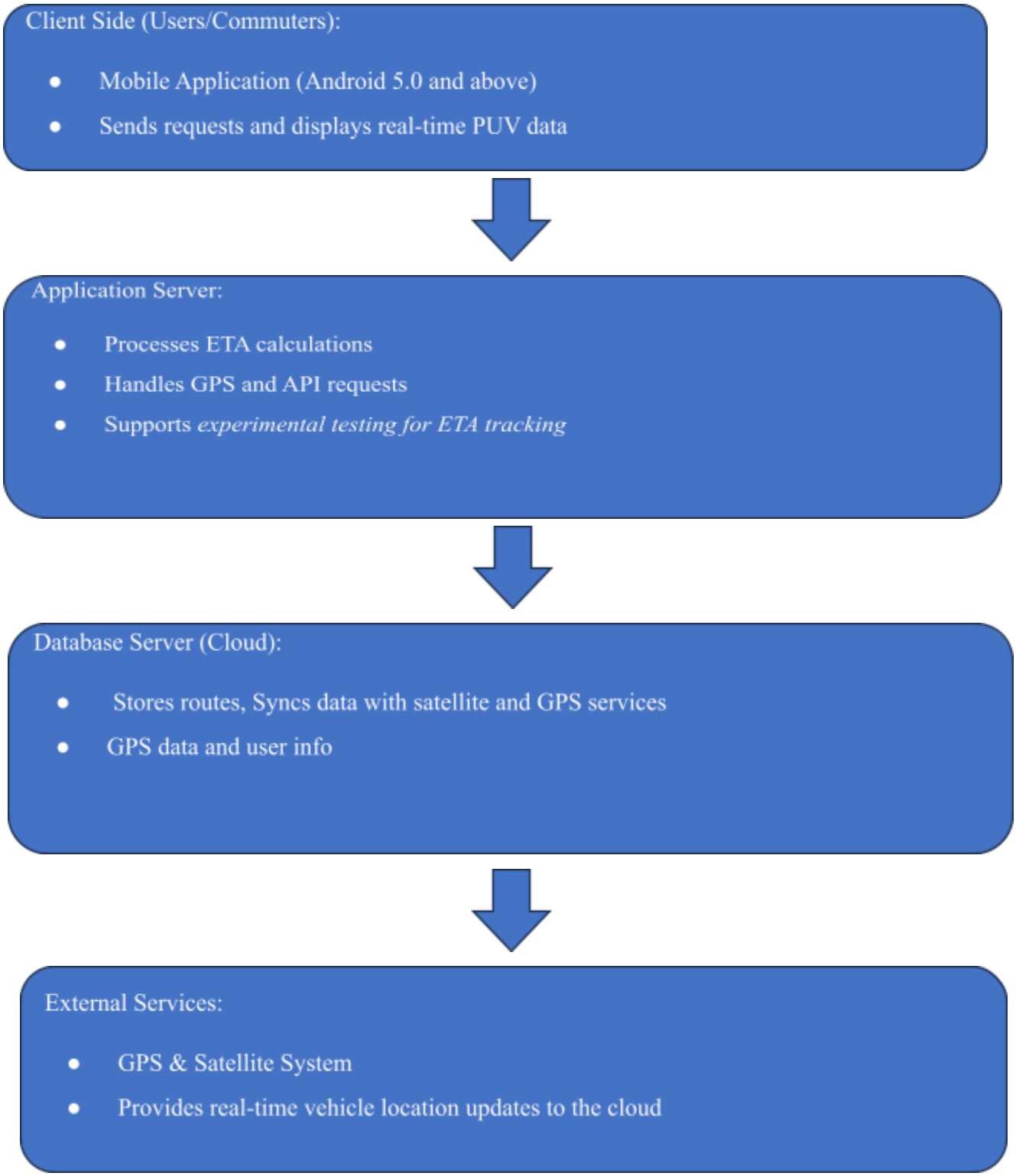

Figure 3. Three-tier client-server architecture model

APPENDIX D

RESEARCHERS CURRICULUM VITAE

PERSONAL INFORMATION

Last Name: **ACEREMO**
First Name: **MERY**
Middle Name: **TADEO**
Nickname: **REMI/MERY**
Gender: **FEMALE**
Age: **17**
Birthdate: **APRIL 10, 2008**

EDUCATIONAL BACKGROUND

Senior High School : **OUR LADY OF FATIMA UNIVERSITY**
Inclusive Years: **2024-PRESENT**
Junior High School: **TALA HIGH SCHOOL**
Inclusive Years: **2020-2024**
Elementary: **KING SOLOMON LEARNING HOME INC.**
Inclusive Years: **2014-2020**

EXTRACURRICULAR ACTIVITIES

Name of Organization/Club/Contents/Activities: **GIRL SCOUTS OF THE PHILIPPINES**
Position: **MEMBER**
School Year/Date: **2023-2024**

"Don't rush the process. Good things take time"

MERY T. ACEREMO

STUDENT PRINTED NAME AND SIGNATURE

PERSONAL INFORMATION

Last Name: **ARCE**
First Name: **PRINCESS VIRGIE**
Middle Name: **CALINGAYAN**
Nickname: **ARCE**
Gender: **FEMALE**
Age: **17**
Birthdate: **JANUARY 25, 2008**

EDUCATIONAL BACKGROUND

Senior High School : **OUR LADY OF FATIMA UNIVERSITY**
Inclusive Years: **2024-PRESENT**
Junior High School: **BAGONG SILANGAN HIGH SCHOOL**
Inclusive Years: **2020-2024**
Elementary: **BAGONG SILANGAN ELEMENTARY SCHOOL**
Inclusive Years: **2014-2020**

EXTRACURRICULAR ACTIVITIES

Name of Organization/Club/Contents/Activities: **STEM SOCIETY**
Position: **MANAGEMENT TEAM**
School Year/Date: **2024-2025**

"PRACTICE WILL NEVER BETRAY YOU"

PRINCESS VIRGIE C. ARCE

STUDENT PRINTED NAME AND SIGNATURE

PERSONAL INFORMATION

Last Name: **AWAYAN**
First Name: **LIVINE FAITH**
Middle Name: **CAMINADE**
Nickname: **LIV**
Gender: **FEMALE**
Age: **17**
Birthdate: **DECEMBER 24, 2007**

EDUCATIONAL BACKGROUND

Senior High School : **OUR LADY OF FATIMA UNIVERSITY**
Inclusive Years: **2024 - PRESENT**
Junior High School: **BAGONG SILANGAN HIGH SCHOOL**
Inclusive Years: **2020 - 2024**
Elementary: **GOLDEN VINEYARD SCHOOL**
Inclusive Years: **2014 - 2020**

EXTRACURRICULAR ACTIVITIES

Name of Organization/Club/Contents/Activities: **N/A**
Position: **N/A**
School Year/Date: **N/A**

"watchu think?"

LIVINE FAITH C. AWAYAN

STUDENT PRINTED NAME AND SIGNATURE

PERSONAL INFORMATION

Last Name: **BOY**
First Name: **MARC AARON**
Middle Name: **SUMAYAO**
Nickname: **RON**
Gender: **MALE**
Age: **17**
Birthdate: **MARCH 23, 2008**

EDUCATIONAL BACKGROUND

Senior High School : **OUR LADY OF FATIMA UNIVERSITY**
Inclusive Years: **2024 - PRESENT**
Junior High School: **BATASAN HILLS NATIONAL HIGH SCHOOL**
Inclusive Years: **2020 - 2024**
Elementary: **PRESIDENT CORAZON C. AQUINO ELEMENTARY SCHOOL**
Inclusive Years: **2013 - 2020**

EXTRACURRICULAR ACTIVITIES

Name of Organization/Club/Contents/Activities: **N/A**
Position: **N/A**
School Year/Date: **N/A**

"MISTAKES ARE LEARNING OPPORTUNITIES; DON'T LET ANY GO TO WASTE".

MARC AARON S. BOY

STUDENT PRINTED NAME AND SIGNATURE

PERSONAL INFORMATION

Last Name: **CABIAN**
First Name: **ZYRUS MC IVAN**
Middle Name: **GUPA**
Nickname: **IVAN**
Gender: **MALE**
Age: **17**
Birthdate: **FEBRUARY 5, 2008**

EDUCATIONAL BACKGROUND

Senior High School : **OUR LADY OF FATIMA UNIVERSITY**
Inclusive Years: **2024 PRESENT**
Junior High School: **TAGUMPAY NATIONAL HIGH SCHOOL**
Inclusive Years: **2020-2024**
Elementary: **TAGUMPAY ELEMENTARY SCHOOL**
Inclusive Years: **2014-2020**

EXTRACURRICULAR ACTIVITIES

Name of Organization/Club/Contents/Activities: **N/A**
Position: **N/A**
School Year/Date: **N/A**

"Time is Gold"

ZYRUS MC IVAN G. CABIAN

STUDENT PRINTED NAME AND SIGNATURE

PERSONAL INFORMATION

Last Name: **GAYLON**
First Name: **AMMON DRAKE**
Middle Name: **MACABIDANG**
Nickname: **JAY**
Gender: **MALE**
Age: **17**
Birthdate: **SEPTEMBER 8, 2008**

EDUCATIONAL BACKGROUND

Senior High School : **OUR LADY OF FATIMA UNIVERSITY**
Inclusive Years: **2024-PRESENT**
Junior High School: **BATASAN HILLS NATIONAL HIGH SCHOOL**
Inclusive Years: **2020-2024**
Elementary: **MELENCIO M. CASTELO ELEMENTARY SCHOOL**
Inclusive Years: **2014-2020**

EXTRACURRICULAR ACTIVITIES

Name of Organization/Club/Contents/Activities : **N/A**
Position: **N/A**
School Year/Date: **N/A**

"GOD DID"

AAMON DRAKE M. GAYLON

STUDENT PRINTED NAME AND SIGNATURE

PERSONAL INFORMATION

Last Name: **GOMEZ**
First Name: **SOPHIA MARGARETTE**
Middle Name: **PAJE**
Nickname: **SM/MARGA**
Gender: **FEMALE**
Age: **17**
Birthdate: **MARCH 28, 2008**

EDUCATIONAL BACKGROUND

Senior High School: **OUR LADY OF FATIMA UNIVERSITY – QUEZON CITY**
Inclusive Years: **2024-PRESENT**
Junior High School: **BATASAN HILLS NATIONAL HIGH SCHOOL**
Inclusive Years: **2020-2024**
Elementary:
EDLEN LEARNING INSTITUTE – 2019-2020
SAN DIEGO ELEMENTARY SCHOOL – 2016-2019
SAINT ANTHONY DE PADUA SCHOOL – 2015-2016
ADONAI SCHOOL OF THE KINGDOM – 2014-2015

EXTRACURRICULAR ACTIVITIES

Name of Organization/Club/Contents/Activities: **N/A**
Position: **N/A**
School Year/Date: **N/A**

"In a world where you can be anything, be kind."

SOPHIA MARGARETTE P. GOMEZ

STUDENT PRINTED NAME AND SIGNATURE

PERSONAL INFORMATION

Last Name: **LUTERIA**
First Name: **RAFAEL LAURENCE**
Middle Name: **BRACAMONTE**
Nickname: **RAF**
Gender: **MALE**
Age: **17**
Birthdate: **FEBRUARY 24, 2008**

EDUCATIONAL BACKGROUND

Senior High School : **OUR LADY OF FATIMA UNIVERSITY – QUEZON CITY**
Inclusive Years: **2024-PRESENT**
Junior High School: **BATASAN HILLS NATIONAL HIGH SCHOOL**
Inclusive Years: **2020-2024**
Elementary: **PRES. CORAZON C. AQUINO ELEMENTARY SCHOOL**
Inclusive Years: **2014-2019**

EXTRACURRICULAR ACTIVITIES

Name of Organization/Club/Contents/Activities: **TLE CLUB**
Position: **TLE G10 REPRESENTATIVE**
School Year/Date: **2023-2024**

"EVERYTHING HAPPENS FOR A REASON"

RAFAEL LAURENCE B. LUTERIA

STUDENT PRINTED NAME AND SIGNATURE

PERSONAL INFORMATION

Last Name: **NICERIO**
First Name: **RUZZELL**
Middle Name: **DE CHAVEZ**
Nickname: **LOY**
Gender: **MALE**
Age: **19**
Birthdate: **OCTOBER 8, 2006**

EDUCATIONAL BACKGROUND

Senior High School : **OUR LADY OF FATIMA UNIVERSITY**
Inclusive Years: **2024-PRESENT**
Junior High School: **JUSTICE CECILIA MUÑOZ PALMA HIGH SCHOOL**
Inclusive Years: **2019-2024**
Elementary: **PAYATAS C ELEMENTARY SCHOOL**
Inclusive Years: **2013-2019**

EXTRACURRICULAR ACTIVITIES

Name of Organization/Club/Contents/Activities: **N/A**
Position: **N/A**
School Year/Date: **N/A**

"CARPE DIEM"

RUZZELL DE CHAVEZ NICERIO

STUDENT PRINTED NAME AND SIGNATURE

PERSONAL INFORMATION

Last Name: **PACRIS**
First Name: **JOHN ERVIEN**
Middle Name: **MERLAS**
Nickname: **JOHN JOHN**
Gender: **MALE**
Age:**17**
Birthdate: **MARCH 20, 2008**

EDUCATIONAL BACKGROUND

Senior High School : **OUR LADY OF FATIMA UNIVERSITY**
Inclusive Years: **2024-PRESENT**
Junior High School: **BATASAN HILL NATIONAL HIGH SCHOOL**
Inclusive Years:**2019-2024**
Elementary: **PRES. CORAZON C. AQUINO ELEMENTARY SCHOOL**
Inclusive Years:**2013-2019**

EXTRACURRICULAR ACTIVITIES

Name of Organization/Club/Contents/Activities: **N/A**
Position: **N/A**
School Year/Date: **N/A**

"A true musician plays from the heart"

JOHN ERVIEN M. PACRIS

STUDENT PRINTED NAME AND SIGNATURE

PERSONAL INFORMATION

Last Name: **TAN**
First Name: **JHAMAIRA MHAE**
Middle Name: **PARDILLA**
Nickname: **JAJA**
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Age: **17**
Birthdate: **FEBRUARY 5, 2008**

EDUCATIONAL BACKGROUND

Senior High School : **OUR LADY OF FATIMA UNIVERSITY**
Inclusive Years: **2024-PRESENT**
Junior High School:**BONDOC PENINSULA AGRICULTURAL HIGH SCHOOL**
Inclusive Years: **2019-2024**
Elementary: **MULANAY CENTRAL ELEMENTARY SCHOOL**
Inclusive Years:**2013-2019**

EXTRACURRICULAR ACTIVITIES

Name of Organization/Club/Contents/Activities: **N/A**
Position: **N/A**
School Year/Date: **N/A**

"IF IT SCARES YOU, IT MIGHT BE WORTH IT"

JHAMAIRA MHAE P. TAN

STUDENT PRINTED NAME AND SIGNATURE

PERSONAL INFORMATION

Last Name: **TUBAY**
First Name: **JAMES PATRICK**
Middle Name: **LUMAWAG**
Nickname: **PAMPI**
Gender: **MALE**
Age: **18**
Birthdate: **JULY 14, 2007**

EDUCATIONAL BACKGROUND

Senior High School : **OUR LADY OF FATIMA UNIVERSITY**
Inclusive Years: **2023-PRESENT**
Junior High School: **BATASAN HILLS NATIONAL HIGH SCHOOL**
Inclusive Years: **2019-2023**
Elementary: **PRES. CORAZON C. AQUINO ELEMENTARY SCHOOL**
Inclusive Years: **2012-2018**

EXTRACURRICULAR ACTIVITIES

Name of Organization/Club/Contents/Activities: **N/A**
Position: **N/A**
School Year/Date: **N/A**

"EVERYDAY IS A NEW BEGINNING"

JAMES PATRICK L. TUBAY

STUDENT PRINTED NAME AND SIGNATURE